

Active InferAnt Stream 016.1 ~ Actionable Template-driven Research: Operational Meta-Science

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All right. Hello. Welcome everyone. This is Active Inference stream number 16.1. It's December 2nd, 2025. I'm going to start the stream with a GitHub push. Everything is on the GitHub repo docology template. We will reload that. And we're live. Looking forward to your comments and any ideas or questions in the chat. Here is where we are going today. We are going to a test-driven development configurable end to end test suite operation data analysis and visualization validation steps and PDF rendering including with citations and clickable links and equations formatted as a scientific PDF as well as a variety of other fun formats including HTML ML files that can be opened for the web and slides or conversion to slides. That's the PDF rendering functionality. There is a LLM review output section that um provides review and improvements and suggestions for the manuscript in English. there's a translation uh output using LLM locally to translate. Then there are literature database search and through APIs and PDF download and local LLM summarization which we're going to use. So without getting too far ahead of ourself, let's go to the agenda. Okay, the title of the stream is actionable templated driven research, which is going to get at the core of what this template repo is, operational meta-scientific arts and crafts, which I guess is like the genre. The stream is live on YouTube on the active inference institute channel. This will be a live stream interactive walkth through and we can also do some improvements and some uh other random stuff. See where it goes. Greetings up cycle. Good to see you. Tools. I'm using a Mac computer. Cursor 2.146 with mostly the auto and the OP recently released available at a discount opus 4.5 and locally O Lama will be using Llama 3 and Qwen type models. Okay, here in this we can look at this on GitHub. There's a few big themes that are going to come up and I'm sure many more. This is just a first step. There's going to be a big focus on end toend configurability and ease of reproducibility. There's going to be the usage of a lot of thin orchestrators in articulated reference which means that the modular flexible methods are underlying in a kind of single composable toolkit uh set of affordances and orchestration happens separated from those tools

which gives good ability to express different workflows and swap out different tool implementations. There's going to be a two layer model to this repo overall, which I might as well describe here. The project folder is a self-contained and is intended to be at the project level, like the specific research paper that's going to get generated. And this is going to be a testdriven package alongside a manuscript expressed in these modular markdown files which are undergoing this internal coherence test suite which could be zero tests or you could use it extensively with the SRC code orchestrated through tests [snorts] and scripts which in the context of a certain formatting for the manuscripts. in markdown. Now pop out of the project is going to be utilized by the template repo using the infrastructure modules and the scripts in the infrastructure and ultimately this top level run.sh at the top level which is going to pop up this main menu where we can run the tests, we could render the manuscript, we could run the scientific review. We'll just start that one up now. So here we already uh in this I have these tabs here is one process going with the local LOM summarizing papers that were downloaded from a literature search. Um we'll look at that step. Here is the output of the translation into Chinese, Hindi, Russian. Here's the manuscript rendering which uh includes the full test suite test suite within the project level of the SRC here with respect to these tests within the project and then the infrastructure level tests which are in the top level test folder. So double test suite then running the scientific analysis which is the SRC orchestration within the project. So here it's like a template paper about optimization methods. Then section by section going through the manuscript markdowns that was the analysis and scientific figure generation. Then there's discovery of manuscript sections and for each manuscript section they're generated into the PDF, the HTML, any other formats. Then the PDFs are concatenated to make project combined. There's validation of the subtypes. The subtypes are copied from output within project which is intended to be a kind of monopilletic standalone kind of briefcase and then it gets copied for convenience to the top level so we can always have the most recent version of the top level just as a convenience. After copying those outputs, there's the local LLM scientific review. And then here is the um warm-up executive summary. And this is is going to um that is here. LLM output project output LLM combine review improvement suggestions. Um beautiful hallucinations and all. Uh, so here is what that looks like when it's running. This is the English scientific review of the current manuscript. This one is on the translation step. We already have them, so I'll just cancel it. And here's the summarization. So this is kind of the other branch. So you can for this repo you can use them together as I hope we will do so in the coming minutes. Um or you could just you could just use the um keyword search and PDF

download summarization pipeline which looks like this. We'll make lit search dot slash run literature search and PDF download 7. This is going to search archive and semantic scholar for papers. Download PDFs, add them to the bib library, generate AI summary, and you can set how many parallel workers. We're going to do um 30 papers per keyword. Search keywords epistemic foraging. Um, resume previous run. No, start a fresh run. So, this is going to search archive for Epistemic Forging. Um, found 30 results, 58 dduplicated from Semantic Scholar and Archive. The um literature output is saved into a top level folder called literature. The references.bib is kind of overleaf latex style bibliography and the PDFs are downloaded into PDFs folder. Then the summaries are made by the same name in the summaries folder and we're going to use that too. Very fun effect I believe and that's kind of chugging away here. So especially with a possibly slightly better naming scheme or bibliographic reference system there could be calculation as I know some teams at the institute and other teams have done to make sort of all of archive scale summarization or work with embeddings other methods but the overall spirit here is this sort of visible craft arts and craft script style that it it it sits in a nice space in in morphospace for programs. Um we will use cursor as a pseudo rag to work with these summaries as context to write researchbased papers that are going to transform the package inside of project and the manuscript. And then we're going to bounce this off each other and get grounded interpretable step by step. Could be entirely manual or could be entirely AI orchestrated with some end toend literature based augmented research multi-channel repo versioning. So let's go to some of the more themes. Uh yeah, two-layer science is the top level of the repo with the infrastructure which is kind of like the oven mitts and then there's project which is like the dish coming out of the oven. So the two-layer science is the meta-cience tools for operations in the repo of template and then project could be plain text markdown for a purely pros philosophy paper or there could be a simple coin flip and the insertion of the outputs of the coin flip or a visualization based upon data for a philosophy paper or it could be extremely technical could modularly compose all these kinds of reference across within sections, papers, all these different kinds of things. By knowing that we're going to use that state space or want to have those kinds of literature expression in making a project pipeline that could be more software with sort of sloughing off documentation or it could be more manual pros and I never entered into it. uh modular minds within and across nested blankets. What kinds of compositionality for agent tools? We can add a MCP or other tool layer. There's already a CLI. uh meta-cience, open science, philosophy, second order science, all these kinds of questions about tool building uh eyides wide open as opposed to the

alternative using even cursor other kilo code other um code augmented approaches and just more generally tools to think with and sense making and personal collective knowledge management how that's related to file versioning. this idea that is big in the DI and meta science uh spaces with version controlling and some of the work with claims graphs and different ways to parse and connect some of the entity reference and semantics and different granularities of analyzing and integrating research papers. uh pseudo make me a manuscript basically role-based access including total root access to research projects for end to end configurable reproducibility including the PDF rendering exactly a certain way and blurring the line using formal ALO reference as well as cursor shared contextualization blurring the line with command, code, knowledge base, all these different kinds of synthetic intelligence semantics which is sort of like the pragmatics and semantics within the augmented IDE which can run code as well as use these softer language affordances. Um, a lot of ways to use the modularity of the infrastructure level and the project level to make packages that are conceivably simpler as well as possibly just different or more modular by adding a module to either the infrastructure level or within the project level, which we can do during this stream. and exploring I think a very creative tension with industrialization and regularization normalization automation. There's going to be some areas to to explore about preparing one's workspace and operations in the niche and sort of to carve one spoon with a knife out of wood. It might be a different startup time or tool need rather than a wooden spoon or a metal spoon factory. But these are all just these different morphologies, different energy gradients, information gradients of production. And so what that looks like in a sort of cognitive semantic information, niche modification, multi-agentic setting. the potter's wheel. Censoring projects before shaping them and making small changes that radiate out through natural forces and and these defined flows. The apprentices advantage learning by structured doing so supporting a niche where a sidefolder like journal or uh field notes. other modular systems can be brought in and and tested in a relatively standalone or in a very deeply integrated way. So with these kinds of niche artifacts as the trace but also with their own activity and informiveness intraactivity in cases as with others we encounter in the niche. So holding up and open that whole continuum from datalike to active like that kind of polycomputing relational feature. All right, let's go to the agenda and meanwhile if you're watching live make a suggestion. I will wait to uh a first suggestion before including that in the research paper that we will re search for and write and update the PDF. Maybe even go all the way to Zenotto at the end of the stream with a literature based. So if somebody has a specific question or uh

topic to research, write it and I'll wait at least a few minutes for that. All right. The way to run this system is to open a terminal in the base layer of the repo. Dot slashrron.sh. This is a single entry point. There's going to be an exit option and eight different kind of partially overlapping functionalities. There's run the infrastructure tests which is going to be let's start with that one in this window. So this is just making sure that all of the modular functionality of the template repo independent of the project whether it's a a pros journal poetry book or whether it's a simulation and data download and integration that is the two-layer model. The infrastructure is like the crust and the project is like the topping of this scientific pizza or or uh analogous. So here it's about to finish. There are 1654 infrastructure tests. So a huge number of many very simple tests and it has about 80% coverage of the codebase of the infrastructure modules which are the the build system which is the PDF composition and validation the core methods of logging and interfacing documentation and relationships between document parts. the literature search methods for download the LLM methods which here are only olama local calls um possibly a vestigial file here the publishing methods which can be greatly extended with nanopubs other kinds of uh onchain off-chain publishing rendering some of the manuscript transformation tools that help go from the markdown with the defined style guide into the properly linked reference and checking for the equations, figures, sections, citations, etc. Um, possible scientific developer functionalities and validation methods that are called on by other package. um every folder where it makes sense. I try to have a both readme a human readable specific accounting of a file function structure etc. as well as an agents.m MD which can describe more similar overlapping more technical sometimes sometimes prioritized information for augmented coding whether using kind of like a codeex cloud code factory AI terminal coding assistant or headless version or using uh cursor like which is doing something similar um under the hood. So that's the infrastructure test function. That was option one. Option two is run project tests. So here these are within project. This is 320 tests but which is still a a vast relative number but just as a starting point for modifying the project package. So this is using these src functions. no modules uh as not to overly bias what the format of the SRC is. Just kind of a flat list of functionalities that these tests within the project again flat covering those file names test and then that is the functions that are going to be used within the package. for example to generate simulation plots. So to simulate visualize check data in general and that's what we're going to modify based upon semantics of literature grounded research summaries. So was option two does this kind of um hey that's just how we've always done it with the high test coverage within the package and at the infrastructure uh level. Okay.

Uh also at the top level of the repo is docs which is documentation about the repo and the two layer system and just a bunch of modular footholds searchable indexable useful epistemic signposting information lot of modularity way ways to improve that. Um so that is the uh that was the second option from run.sh SH which was running the test suite for the project three is the uh rendering of the whole PDF manuscript. Let's just do that one. it um discovers the project runs the generation of the simulation and the visualization by running an orchestrated order of scripts within the project folder. So that the project folder again is kind of like this complete briefcase unfoldable source code orchestraable DNA. Then each of these sections in the project manuscripts folder are going to get generated slides. I will open up a I'm going to open up a window and just so we can look at the um dashboard. One second. Okay, here's the um rendering of the manuscript finished. So, it renders into output and then it gets copied into the top level output Here's the raw data from the simulation. Here are just the figures from the analysis generated by the scripts. Here are the LLM local LLM based. Um that's looking now at option five, the LLM scientific review. in this case in English um combined review. Let's look at these on GitHub. So project output ll project output here's combined review. So for context, the paper that we are reviewing is a sort of appropriately referential example default project title subtitle example by project author on today's date. And it is a red letter absolutely black, white, and red all over document that's composed to a cited linked interactive format, but also these other formats for web. So let's see if this will open and brave. HTML for website use. The slides are saved. So probably some uh manuscript formatting is going to look uh look well structured or not in uh if you write short paragraphs or long paragraphs. Uh it might look good in the slides but not in the HTML. Maybe some of the formatting is a little off. Something like that. Webs, slides, simulation data, the um validated config, parametered reporting, plain text, composable reports, the PDF, [snorts] latte basis, bibliography file, and final linked PDF. So that's and then the translation maybe only goes to top level. Not sure where those went but there we'll we'll we'll run the translation. We'll do so step six. Okay. So now in this rename that terminal to translation here we have summarization happening. So with this step okay kill that one. Right. So that's some of the um Hello Gabriel export. That's some of the manuscript rendering project running project level deliverables within the context of you could just be doing that with all manual writing. Doesn't need any kind of literature. You could just delete the bibliography or have a null bibliography. So now um let's look at the step seven and not eight or literature search PDF download. So here uh I'll see a few more minutes if anyone has a specific topic that they want to suggest for which topic to research uh

especially if there's archive papers on it. And then we will make a branch of the template repo and go end to end on making that well formatted PDF with literature grounding. But here's the literature search PDF download component. We'll just do 10 papers per keyword expected free energy interospection electromagnetic. We don't want to resume the existing run. just picked a few keywords could be improvements to the um API call literature database methods. So just 10 results from archive. I don't think I entered any email or API for semantic scholar. So hence probably why the rate limit is getting hit there. So it's going to search these three separately and maybe just doing archive for certain topics would be would be a little faster. Let's keep watching. some there's several there's about 600 PDFs that I've already downloaded that are in git ignore as a folder so they don't go to the uh GitHub and then if the PDF of that exact compos name author year word if if it's not found it tries to download For some of the papers, the current download methods work, but not for all. So, uh, but for for many, let's see, 14 were already downloaded for those topics. 20 were newly downloaded, six failed. So after um downloading the terms, it tries to process those uh PDFs. So let's look at K 2023. So this is probably something from the in uh epistemic foraging or something. We'll see. So okay top infrastructure level literature PDF KO U W 2023 information seeking okay so this is by voter cow of institute discord fame information seeking polomial narc model predictive control through expected free energy minimization. So this is a archive paper from about two years ago that right now with the llama 3 gradient language model via o lama is getting summarized. So that's going to go into we'll pin this one that is going to output in the coming seconds to minutes a summarization in the summaries with the same with the same file name. So, let's just look at the PDF, see what it's about, see what the uh what the what what what the list brought in. We'll just read the abstract for epistemic value. We propose an adaptive model predictive controller that balances driving the system to a goal state and seeking system observations that are informative with respect to the parameters of a nonlinear autoregressive exogenous model. The controller's objective function is derived from an expected free energy functional and contains information theoretic terms expressing uncertainty over model parameters and output predictions. Experiments illustrate how parameter uncertainty affects the control objective and evaluate the proposed controller for the pendulum swing up task. And that's getting summarized here. Greetings paradoxical AP. So this uh here's the translation outputs [snorts] that is in template project output lm translation. Maybe not. Okay. Say love you. Okay. So now here 456word summary of this cow paper was written that's going to be in literature summaries how 2023 that was previously not there. So this is just using the

O Lama local LLM to summarize the uh paper that we just read. Okay. So those are the two branches of functionality that are at the infrastructure level that enable different kinds of project level work. So let's kill the processes and then take the project in a specific applied direction. check in here. I'm going to make a new uh branch that's going to use the tools in their functioning state for main. And this is going to be just live branch. if someone has a research question or topic that they want to write in the live chat then I I will prioritize that just to uh give a fun direction injection of of randomness. But what's going to happen and we can start writing out the prompt here already is um we are going to transform using cursor as a planner and IDE transform the project folders source code manuscript and summarization into a specific research project. while we use the literature search methods to search archive for research works that get summarized on that topic. Then we're going to use the summarization of some key archive works in our context window to update the manuscript and site those archive papers properly. So that we're going to go using cursor within this context of end-to-end reproducible configurable PDFs. We're gonna go from this template placeholder PDF into a uh researchbased documentbased open-source software packaged unitriven development improvable modular system with actual reference to citations that are of the topic that that uh someone can suggest. But we'll start the prompt and just wait like one more minute if anyone come does a prompt otherwise I'll make some random question. Okay, for the following topics, make plan to make any and all changes to only project. All right. Gabriel export wrote something about containment theory. Okay. I can kind of guess what it is, but I don't know what it is. So, um, we will do actionable theory, met literature, meta analysis, and some question. Add more details if you want about containment theory. But first, I'm going to start the literature review for containment theory. So, run.sh. Step seven, literature search PDF download. We're going to do 50 papers, search keywords. Okay. Containment theory, comma, and Gabriel wrote alternative to set theory. Set theory. All right. So, this is going to you can imagine letting this run overnight. So, we'll just look at uh this. This is actually going pretty fast. So that is going to download the PDFs for the search that we just made for 157 citations and add them to our bibliography. So that's happening. Now we're going to transform the project folder to be a project about containment theory and being an alternative and implications to set theory. we are completely transforming the project folder within the constraints and niche of template slash infrastructure methods as documented and tested. We are transforming completely and only the configuration source testing methods rules etc. to be about I'm writing down what Gabriel export has written in the chat. Containment theory as

alternative to set theory also known as boundary logic or iconic mathematics. comprehensive bibliographic based. This is where we point to the summaries because cursor can access across these folders for real summaries of ref specific comprehensive literature review sections. Technical, analytical and gestalt overview. Unified notations and tables, visualizations, methods, tests. What are the what are the basis implication testable hypotheses and explanations comprehensively. plan to transform all this is just speculative and and uh fun but these can be tuned as well. Transform all sections as and all downstream. This could be just entered in um to a single run method. Okay, here uh using Opus 4.5, which within cursor is available at sonnet 4.5 cost during the first two weeks, which I guess we're still in. Um I should have enough usage to use opus a bunch of times during the remainder of this stream. Um what's happening now as we are on now the live branch. So this is now main is what I forked to make the branch that we're now on. So main is the one that you just check into. And then this is now where we're vectoring off from. We're adding in URLs. Uh we're adding in doing literature to uh get new PDFs and summaries. What formulation of containment theory? Um both since it you how should we handle references? We could connect to existing active inference literature. Let's do it. Mathematical formalism full spectrum. So the plan mode in cursor which uh some or most but not all of the models have but opus 4.5 does. engages in a preliminary round of context iteration and refinement tool use internally. And we'll then write a plan which is then sequentially with some um uh very interestingly controlled kind of uh stubbornness is. So here here's what the plan looks like. We're going to transform project into rigorous research monograph on containment theory. An alternative to set theory based on G. Spencer Brown's laws of form. Let's just see if this top level point is like already way off the map. We'll let it finish writing it. Um, oh, containment theory replaces set membership with spatial containment. Okay, let's just let's let it cook while we search. Little pre-dinner reading. Gabriel in the chat and this distinction network. So that that was also that was uh with without containment theory summaries in the sum without containment theory papers. So here we finished the 157 new papers were downloaded. Now those are going to get iteratively processed by Olama with a llama uh lama gradient local llm call ye 1997 variational. Let's just look at the PDF. Ye variational. So this is this is going to start making a summary of this 1997 paper. So you can imagine that with a more uh powerful compute which might be done one or a few times at a few different lengths or with uh old PDFs but that's done once per paper. So that is going to be plugging away. It's not going to come into use probably in this stream because uh a few only a few will come into play. We'll we'll see. Maybe we'll just let it run and

we'll get like 10 summaries. Then we could ask, okay, would these 10 new specific papers add to the paper to make it specifically in relation to this citation, that citation, that one? Okay, so while we're thinking about this kind of meta meta review literature summarization paradigm, the containment theory plan is underway. So here within So here's the agent implementing that plan. So it started by this is all heavily signposted in the documentation and the cursor rules and and everything. So the config file of within the project was changed to containment theory boundary logic as alternative foundation to set theory name. I'm going to just do stream and I'll put real email. Uh some tests config.yaml the full uh so here's configurable title here. Here it'll just be example author information, publication, keywords, metadata, and which languages we want to translate the uh abstract to in that language step. Then in the SRC folder, it next implemented a 525 524 line code boundary logic forms. This module implements the fundamental structures of G. Spencer Brown's laws of form 1969 providing a computational foundation for boundary logic as an alternative to traditional set theoretic foundation. The primary distinction or mark creates inside outside relationships replacing set membership with spatial containment possibly like a sort of discretization topology as in duality with a continuous information geometry. Key concepts form a recursive structure of nested boundaries. very relevant Mark the primary distinction creating inside outside geometry um the bubbles trilogy by slaughter dyke of course synergetics tetrahedral as minimal space enclosing system isotropic vector matrix void empty space the absence of distinction cross the mark state true so that's kind of intraactivity, a fundamental relationality, unity and duality, uh interactionism. And to what extent this is true, someone more specifically can tie to the literature. Um or we could have primed with the PDFs of these textbooks and papers specifically. So, let's look up. Let's start another literature search. Run literature search PDF download 30 papers. G Spencer Brown. So then uh this is going to continue. But then you can imagine that in let's say um 10 to 15 minutes we will be able to say okay update containment methods in light of the formalisms exactly like in these papers and make sure you cite Okay. So, we'll just we'll quit that process and just focus on the focus on just the search for G. Spencer Brown. So that's the only literature thread. Now we have still the SRC methods getting written by the opus 4.5 agents in the project of pivoting project folder to be about this research question. So meanwhile just to get some information. So in Yep. So this one this is lit manuscript end to end the full pipeline. this is going to first erase the output directories in the project level and at the repo level. Set up the environment and check all the needed package dependencies at the infrastructure and at the project level. First run the infrastructure test suite.

This is just to get early warning before getting at all into the package. Making sure that all these like 1600 tests at the repo level are very strong. Now we're going into the test suite. Now the test suite failed because it didn't hit 70% coverage. That's another sort of early warning flag just to exit. Um so that there's end to end viability. Um it I'm sure there are as in in this moment excursions from that end to end but it's kind of like sobbing breathing. uh seems good to keep it on the rails and then if you want to change some aspect or remove some aspect that can be easily done. But there's something nice about that run halting at uh uh ordering which is configurable um for that one option. But you can run other options like if you don't want to render the manuscript, if you just want to do literature search, then the um the fact that the manuscript won't render render doesn't halt the literature search download summarization. Still, we have the agent plugging away the code. we can look at the plan and basically see if the next strategy okay so after so first the source module so that's where what's happening diagrams py is happening looks like that was not included in the original plan but I guess hey plans We'll see after this diagrams finishes maybe that will finish this first section on the source modules and then there will be these rewrites of the manuscript uh sections. that was there was the the thought there. Quote thought good progress. Now I need to update the init.py then create the test suite. Let me continue. So completing the test suite confirming the functionality of the SRC that just happened will raise our test coverage to 70% in a few minutes. So we will get past this threshold. So this that will be a good pace. Meanwhile, let's check back in containment. All right. So, we're Let's look at a containment theory summary. Let's see. There could have been citations in different fields that that do or don't have primary relevance to what we're studying here. or what are we studying here? But let's see. All right, we have [snorts] expected performance of the Atlas experiment. The Atlas detector. Wow, how relevant. How cool. [sighs] Many author paper title instability coupling P modes and G modes within neutron stars on GW170817. B particle mu decay high energy neutrinos. Not sure if these are the papers that we are looking for but again it's a uh example of what can be curated in this repo by by anyone who would like to explore. Meanwhile let us remember that we have 54% with 320. Let's run it again. So we have 54% 320 going to run the whole pipeline. So as we made no changes to 54 320 we will expect homeostatically that the infrastructure 1600 tests fully pass and and some of the slower ones could be skipped. some of the LLM methods are are skipped because they're they're long generation or maybe somebody doesn't even want to use the local LLM methods at all which are not required for the manuscript being generated but those passed. So now we move into the okay then right off the bat we get we hit now the

test errors within within the package. So, just to kind of keep things on the rails, going to reync here are the current results of the Replan it. >> [snorts] >> Okay. So now the plan is being updated. It's about pi test. So some files are going to be updated at these specific planned locations. Then we will get to the manuscript getting written. Oh, J1 J2. That's so funny. Here, let's do a little J1 J2 lore. Those are two of the lineages that need each other as it were of the pogamir southwest US species concept that I did my PhD dissertation in in this region. So it's just funny because it was a J1 and J2 population. So let's see what J1 and J2 do here. opus 4.5 improving the intra package test suite then once that happens we will get through that next step of running the manuscript end to end because we kind of don't even want to look at the manuscript if we don't even have the tests functional. Um, and once we do this initial pivot, which is still in in progress, then we'll be kind of switched over all the way to this other vector. But now we have these small edits which go relatively fast phase one. Updating the source files. Now updating the imports. Definitely a fun way to see a lot of sample size and learn a lot about coding and Python. Seeing it all even when it's not it's not uh 100% but it is vast and accurate in many ways. not always in the only way or in the specific way uh unspecified but some of these coding practices are are really interesting to see just flexible at this level and with um opus and a lot of other models right now. All right, [sighs] back to the agenda while we give it a little time. Just, you know, give it like 20, 30 minutes. We've only given it a few. All right, so we uh ran run.sh SH talked about these two kind of branches of project level test orchestration analysis rendering and then on the literature search branch the um tools to do research summarization which then we can drag in and you can say okay now take this file into account when writing this manuscript which is going to be one of our next steps. um project from scratch. The topic was brought up to do this containment theory topic and we're able to touch upon in in this direct uh show and or tell strategy with live stream interactivity. A lot of these different aspects of meta science or just science itself like structure and function and different ways to think about the science ecosystem as different kinds of or organizing processes processes with different information flows taking a systems view also firsterson view as a navigator epistemic forger in that space and uh just modernity and and it all motivating this is the ability to increase the quality and quantity and the reproducibility and integrity of scientific work and bridge and upskill and and upniche and leave it better than we started with respect to open science and all that there's this two-layer architecture where infrastructure literature docs test, scripts, and run.sh all apply to this infrastructural kind of oven glove handler level. And then project is intended to be this

standalone. We'll just say um now just get to focusing on the manuscript updating can do code later keeping in mind using the very most relevant specific citations of. So here's the project kind of reaching outside of its own folder and then using cursor like rag to softly access the information geometry of summaries or or there could be another rag call or an o lama call to this folder. Um then use the bibliographic information and bibliographic format of the enforced by and and acted upon by the infrastructure methods within the output within the project folder manuscript. So then we'll have our library PDF downloaded papers other documents summarized however we uh want to LLM summarize them done done by a community for thousands of papers or done by one person for even even thousands of papers as it's going through dozens just on this laptop. top within a end to end versionability setting. So now we're going to run the full pipeline. We expect to have the infrastructure tests pass and then the package tests fail. Comprehensively fix all project scale methods and tests as will pass 70% coverage. We expect the infrastructure 1600 to proceed. We expect to surface some of the errors uh from the SRC project SRC in the coming few seconds. There we go. Two errors. Make sure that everything is synced. Now with these two agents vectors which can kind of co-coordinate through the sync vector embedding of in kind of coming up for error I I think either at the prompt user inter interface level or in between these smaller steps. Um, now we have the tests which we want to get to 70% so then we can get to the manuscript rendering and the analysis and service the next level of of errors. Now we're already writing the paper. [snorts] So building on G. Spencer Brown's laws of form 1969. We develop a complete computational framework for boundary logic that demonstrates its equivalence to boolean algebra while offering distinct advantage in parsimony, geometric intuition, and handling of self- reference. This work establishes containment theory as a viable alternative foundation for discrete mathematics with complete computational verification of its theoretical claims and open-source implementation for further investigation. So that sounds like something that uh would require that kind of theorem proving environment. Uh we do we will find out whether that check written by the abstract is going to be upheld by the analysis. but we're chugging away. Or we could tone back whether it's a formal proof and just feed in say we want these seven papers to be included in a more of a literature metaanalysis paper or or maybe we'll see what this literature review um document looks like. Meanwhile, we have the fixing of test errors. Let's just more generally run the literature summarization step just because we're not using the LLM otherwise um generate summaries for download PDFs. So first that's going to try to redownload missing PDFs. We have 922 papers. 167 need download. 583 need summaries. So

we'll just let that run. We have the now it's iteratively running and reingesting the project level test suite which is going to um give it a few rounds of being able to feed in the errors in the failing test. So we can also peep that button and look in sometimes you can't see it because of the piping command but um we can see it kind of ticking away here. So that's within this cursor infinity window here it's testing the algebra complexity. Then over on this one, we still have the references. Here are the Wow. Calfman CS Pierce. This is awesome. Say excellent. Ensure that one of the supplemental sections is a comprehensive, vast, formal, concise representation and grounding in with the history of North American pragmatism and neo materialism. just to sort of mix it up. So, we'll see. Will it finish so then accept this cued message or will it finish a series more methods and then we'll just send that one down just so we kind of get through to it. Okay, test suite is ongoing. run sh again run full pipeline or we could just do run sh to run project tests. so I'm going to stop this one. just to give it a re context comprehensively fix all the project tests. Okay, hope I can finish this in the next few minutes. Let's just review the okay we'll look at that for sure. So we have project tests developing plan in progress. Here is the philosophical esso. So introduction G. Spencer Brown structure. Supplement one is like a technical code. Supplement two is axiom verification. So this this can be pure equation play with with vast painting. That's the kind of recipes that you may or may not be able to to come through on and or can be injected or confirmed equivalent to formal theorem proving and/or numerical simulation steps in the SRC of the project. Then we have this new so this is the okay CS Paris developed existential graphs a diagrammatic logic that anticipates Spencer Brown's calculus in fundamental ways the connection is not merely superficial but structural very cool very knowledge graph pled very computational semantics. William James, John Dwey to exist is to be in a situation which is in the technical direction the kind of minimum system start with universe two eyes open approach of synergetics. It's Dean's sort of zoom in zoom out. It on the performative side is more like situationism or all these other kind of math art or uh art related aesthetics. Alfred North Whitehead the category of the ultimate Karen Bar. I know from the paper of Mao and Ali that that Barad's agential realism is related virtual and the actual deloo onto power core commitments. So this is very long the tour. So even to the limited Donna Heroay even knowing this in a limited capacity that seems to be Richard Rory lot of crossover there. Right. So we have documentation getting updated at the level of the project to reorient it towards being stably for this containment theory approach. Here the thinking happening with the test. We'll just say get going. But that part could range from this is sort of now the lower bar is that

kind of thinking that was just being quote shown or quote thinking that was being shown and that could include MCP or other orchestrated calls to doing more specific theorem proving in methods. ensure all methods and tests are. This is going to be a little bit of uh these sometimes can can uh interfere with each other's work with fixing different tests, but we'll just run these two because they'll probably be staggered in how they run tests. Um so as long as we keep refreshing the context, we should be good there. and just let us check how many run project tests. So that's what it's running that in this agentic tab. [sighs] All right. So, so 16 failed and and we will not move forward until those. We could also just ask, is there a way to accept failing tests and still proceed? So targeting test failure by test failure. We're in the most proximal semantic the bibliography and the claims in the manuscript for the SRC. So that's the sort of standalone dual channel semantics is the formal and the pros methods as well as a secondary documentation layer to to connect any needed dots like like just sort of lab materials to make an integrated unity and coherent functionality at the project level the briefcase level and then we're in the document factory with the infrastructural layer And so partially just to maintain that end to end functionality whatever pattern that name is a high threshold is kept for the tests just so that there's minimal drift away into like a nonfunctional regime which might come through a uh a alley into a new zone of of improved functionality, but then that should be branched and then brought back in modularly. So on this stream uh we may only in the next few minutes uh get to that PDF rendering, but suffice to say that I I can let it run while I continue doing the rest of uh work and play and arts and crafts today and just upload the the PDF which would go to the outputs PDF folder. and uh Gabriel export. I hope that that this gives some some idea of what the system can do for everyone. So I appreciate the suggestion. We have abstract the introduction that we looked at the methods theorems experimental computational kind of irreducible complexity. Ruleadian acknowledgements appendix. Yeah, I would love to see the output too. I want to stay a few minutes just to get there. appendix symbol glossary and that that can be checked between the names of the variables in the SRC and then within and across manuscript sections references is a placeholder for the bib. Yeah. Wow. Thought for 4 minutes and 40th. Okay. So, wow. let's run it again and see how many failing tests. 16 488. Run project tests. This is the most recent test run. Fix everything given the current state. Yeah. Just reync it. Let the other one also continue to run, but let this one do a little plan. Actually, stop that one just so that we're down to just a few failed six. So, finish these six tests. That might take a minute to skip through. So meanwhile comprehensively review all the manuscript sections. It is making maximal

compliant real use of all relevant citations from literature. Just let it run. We'll just say mark them and remove the block for now. We will fix later just so that we get to the manuscript. Okay. So now we have let's Wow. Looking at the six remaining failures, they all relate to a single issue. The excluded middle. How deep? Okay. Adding the decorators so that we can proceed hopefully through those Now comprehensively proceed to one by one fix and confirm all those tests. So now that's going to fix the six. So that might fix these six once these. Okay, now we will do run.sh. We'll just we'll just run the full pipeline and just hope that this is going to work. Meanwhile, open up the cursor. Okay, it's not going to show us until it finishes. So here we have close that here was from that background process of getting the PDFs. Looks like we didn't get any PDFs, but it's about to Oh, maybe. But then it's about to uh enter into the summarization phase of these 167. Right. Meanwhile, all right, here we go. So now we got further in. So now we're generating the slides. So now we're in project output PDF. So now it's here are the slides. Now here's introduction. So this is the autogenerated slides notation. Here's the experimental results TOC tax. methodology slides, then PDF paper. Boom. So red red letter scarlet letter text play rubrication document structure there's a reference error but these get flagged so you you can fix and then also the iterative feedback is okay well equation crossing is not proper there's some visualizations some tables there's the citations so very cool end of stream I am going to end it there I hope people enjoy and find it. Well, till next time.